

Kevin Warsh is right to say, what are we doing with this balance sheet? Why can't we ever shrink it ever again? Are we stuck forever with the ample reserves regime? Those are really good questions. And in the past, the recent past for the Fed has done these policy reviews. It has absolutely failed to ask really hard questions about what its tools are and whether

they're working. I think it's important that we're having that process. Welcome to Top Traders Unplugged. In markets, success doesn't come from predicting what happens next. It comes from being prepared for what you can't predict. In each episode, we go deep with some of the world's most thoughtful minds in investing,

economics, and beyond to understand how they think, how they prepare, and how they decide, and the experiences that shaped how they see the world. No noise, no shortcuts, just real conversations to help you think better and invest with confidence. Welcome back to Top Traders Unplugged. My name is Alan Dunne. And today, I'm delighted to be joined by Brendan Greeley.

Brendan is the author of a new book called The Almighty Dollar, 500 Years of the World's Most Powerful Money. He is currently completing a PhD at Princeton University, specializing in the history of money and finance. He was previously a financial journalist for 20 years at Bloomberg Business Week and the Economist, and he was US economics editor at the FT, and he continues to write a column

for the FT. Brendan, great to see you. Great to have you on. How are you doing? Good morning. Thanks for having me in.

Well, I mentioned you've written a new book called The Almighty Dollar, which is really fascinating history of the dollar, and we'll definitely get into that. But we do like to get a sense of our guests' background and how they got interested in economics, finance, money, all of that. So how did you stumble into economics? I did, in fact, stumble into it.

That's the exact right verb. I was a German literature major in college, a completely useless degree, which prepared me to make very little money as

a German translator. And then I became a journalist, and I was right after the financial crisis, I realized that there was a niche in my own publication at the time I was at Business Week, where nobody was covering economic theory, and there was a lot going on.

There was a lot of things that had been missed, as we know, in the financial crisis by the economists who were in charge of America's economy. There's an absolute iron grip that economists have over policy in America. They were having a think, or having a come to Jesus over what they had missed and what needed to be done. That was a lot of fun reporting.

I got to talk to people at the Fed about how you set up a macroprudential crisis facility, meaning how do you set up a group of people within the Fed that worry about a financial crisis that continues to be relevant when there has been no financial crisis in the last 10 years? I talked to the Fed also about figuring out how to integrate money and banking back into economic models.

This is something that I think is a real challenge right now for economists still today, which is they have these black box models of you put in employment and interest rates on one end and get inflation out the other or any of those three variables. There's lots of things that can go into that. They talk about forward guidance in various ways in which the Fed has of influencing that model.

It doesn't really want to look about at money and banking. We know the money we have in a developed industrial economy comes from banks. When banks make new loans, those new loans create new deposits. Those new deposits are new dollars or new euros. That seems like a really important part of the economy, and particularly for the Fed, which is a bank, which is in charge of our money, to understand that and model that.

They just don't. Since the early 1980s, they've really abandoned the approach of thinking about the production of money, different kinds of money, what's M1, what's M2, how many M's are there. They just don't think about that anyway. That's not the way economists think. They've really walked away from that approach.

They walked away from that approach because they couldn't figure out how to create one big model of the entire economy using the money supply. I think they missed something when they walked away from that, which there are lots of little components of the money supply. There are different kinds of M's. The Fed doesn't really think about the different kinds of M's anymore.

I did that kind of reporting, and I thought that was fascinating. I eventually ended up being the Fed reporter at Business Week and Bloomberg, and then eventually from the Financial Times as well. I love looking at the Fed's balance sheet. I love looking at the Fed's tools. The Fed doesn't really like to talk about its tools.

It likes to talk about policy. Are we going to ease or tighten? That stuff's important, but how they ease and how they tighten is really important as well. I ended up being in a position to write this book because I was doing things like looking at QE and trying to figure out how it worked.

It's actually really hard to figure out how QE works and whether it works. I'm not completely certain that the Fed is sure that it knows how QE works. I think sometimes the Fed thinks, well, as long as we don't do anything overtly political, then we can't get in trouble. Buying a bunch of treasuries to them feels like it's not overtly political. It doesn't help one party or the other.

That's a thing that they can do. Just because you can do something doesn't mean you should do it because it might not work. I'm not convinced, for example, that QE is doing anything other than raising the value of existing assets, which is an intensely political act. It was a result of that reporting that put me in a frame of mind to really think hard

about how does the Fed work? How do we actually manufacture money in America? I was much more interested in all of those tools than I was in predicting the Fed's reaction function to various economic stimuli. I think now that I've gone back to school to study the history of finance, I'm completely ruined as a Fed reporter because one of your jobs as a Fed reporter is to sit in that room

when they give the press conference and ask very important questions of the Fed share. I think at this point, if it were my turn to ask a question, I would just say, what are we even doing here? I don't think that would be a very productive question for my employer or for the Fed. That's how I arrived at where I am today, is thinking as structurally about how the Fed works, how banks create money, how those things work together, and how that's not

in any way reflected in academic research or federal monetary policy in America. Interesting. You've touched on the fact that you're doing your PhD at Princeton at the moment on finance, money, history. I think you're four or five years into it now, so has that experience changed your perspective? Obviously, it's changed your perspective a little bit on your profession, it sounds like,

but on economics more generally, or what insights would you say have come from that intensive period of review of economic history? First of all, Alan, it's just really fun to read a bunch of books for a while and check out of what you're doing and just learn a bunch of stuff. I was doing a project with a 27-year-old grad student, my cohort, they're all 20 years younger than I am, and at one point he said, can you believe you're doing homework again?

Then I said, I can't believe I get to do homework like that, this is fun, this is great. I think that I was exposed to a couple of historians who've really helped me think both about how I write history, but how I think about what the Fed does today. Importantly, my advisor, Harold James, really thinks about finance. He's a financial historian, not an economic historian, and that may sound like a distinction without a difference, but the difference is he really forced me to think structurally

about how all of these historic devices worked. How did a bill of exchange actually work? It's actually really hard to understand a bill of exchange. For me, I had to get pieces of paper, write them out, and then pass them back and forth across the table. Now, when I teach students how bills of exchange worked, this is this technology that's been

around since the 14th century and endured until the 20th century. It's a very difficult challenge to actually send value across an ocean from one place to another. I would actually divide up my students into two halves of the room. There would be a Florence side of the room and a London side of the room, and I had them all write out bills of exchange and ship them across the room back and forth to figure out

like, what was it? How do you write it? What do you sell it for? What's on it? That's sort of work actually forcing yourself to structurally understand things, rather than say there were bills of exchange, they moved value.

Has really helped me think about how money works now. There's another historian who's been really influential. Her name is Rebecca Spang. She wrote a beautiful book that I would recommend to anyone called *Stuff and Money in the Time of the French Revolution*. She asked really difficult questions about how money literally worked.

When you say somebody in Paris paid for something, what does that mean? Literally, what did they hand across the barrel head? How did they pay for it? Was it coins? It turns out it wasn't coins. It was a lot of it was just oral credit.

People just remember what you owed them. At some point over the course of the year, when you had cash, you would settle up. That turns out actually to be how money worked in a lot of places. I like that book so much. I got in touch with Rebecca Spang. We talk a lot now.

We both agree that what we work on is something we're calling low finance. How do people pay each other? Turns out there's a lot of ledgers. There's a lot of temporary credit. There's a lot of handwritten promissory notes in history. There are a lot of tokens that are produced by shops that allow you to make small change.

All of this sits underneath of the big finance of bank notes or Federal Reserve notes eventually. There's a long process through which America figured out how it is that we use things to pay each other now. That historical process is really informative. It helps me be pretty disciplined, I

think, in thinking about how do payments work now? I think in America, we're awful at payments.

Europeans complain about that. They come over. They realize that in America, we still write checks. Yesterday, I was in the archives of the American Numismatic Society. I found a check that was written against the cashier of the Bank of North America in 1792. Alan, nothing has changed.

I took a picture of it. I put it online. Nothing about the format of a check has changed. In America, we still write them. We can go into the policy reasons. My argument is we should pay attention to low finance because it's how all of us make

small loans. It's how all of us access liquidity. It's how all of us really, it really governs our daily lives in a way that's incredibly important for policy makers and they just don't think about it. Payments is a backwater at the Fed. It's the assignment you get when you don't get to do the big, fun, important macho macro

stuff. Well, interesting. It is one of the themes of your book as well. You don't call it low finance. I mean, you talk like big money or small money. I think it's a different term for it.

But it is kind of a thread that runs through it. So maybe talk to us about the book. Obviously, you took it on as a project. I think as you got into the project, you probably learned some things that impacted the direction. So what was, I guess, the big surprise? What was the big insight you were trying to illustrate for writing the book?

I think the value, you always have to ask yourself, why am I writing this book and not somebody else? What am I offering here? I think what I offered in the book and what informed my approach is I was learning to be a historian over the course of writing the book, but had worked as a policy reporter and one of the things that I think historians do, particularly about money, maybe about

other things, is that they assume that modernity changed everything. When I was at a conference once and I heard a historian say, we were talking about

small change, say, well, then we got the Federal Reserve and that fixed all that. And I was sitting there as a recent former Federal Reserve reporter thinking, no, we absolutely did not. The Fed didn't.

I mean, there are lots of challenges. The Fed's a great institution, but there are lots of challenges that it has failed to fix. And it allowed me to see that I think that there are continuities between the way money worked in the 1520s when this book starts. And the way it works now in the 2020s where we are today.

Most significant among them, as you bring up, is the idea of big money and little money. So numismatists, people who study coins, know from mint records that there's a fundamental challenge in minting that is as old as minting, which is small coins are more expensive to produce than big coins. And the reason is you have to hammer out a small coin literally until the 19th century, you would stick a die on a stump, the obverse die, and then you would stick the little chunk

of metal on top of it. And then you would stick the reverse die on top of that and you would sit on top of the whole thing and you would whack it with a hammer. That's an industrial process. It's very difficult to do. It takes skilled labor.

Crucially, every coin needed one whack, which meant that per unit of value, large coins are much cheaper to produce than small coins because you have to have many fewer whacks, a lot of less skilled labor. And so that is informed minting policy, honestly, since the first Electrum coins in what we now think of as Greece. And I don't think that challenge has gone away.

So we refer to that as brassage cost. This is the cost of production for minting. We now, of course, don't make money with coins. Coins are a very small part of our insignificant part of our monetary system. We make it through banks. However, bank loans have the exact same challenge as minting.

It's just as expensive to underwrite a small loan as it is to underwrite a large loan. You got the same back office costs, no matter what, but your profits are much higher on the big loan. And for the exact same reasons as mints

prefer to make big coins, banks now prefer to deal with high value depositors, high value borrowers, high value transfers. They're much better profit centers than small banking is a real challenge.

You know, in America, we have a much bigger network of community banks than you do in other countries. But we're still uniquely bad in America at getting people into the financial system, getting them into the banking system. We have a lot of families in America that don't even have bank accounts. And precisely, and the reason why is that banks make it very difficult to be a depositor

and a borrower if you don't have a lot of money. They just don't want to deal with small loans and small deposits. So my argument in the book is the central challenge of money has not changed. Brassage costs are still high for small loans for banks, the way they were for small coins for mints. And I think that those challenges we're still stuck with now.

And so one thread that runs through the book is often the way normal people buy and sell things. So not merchants, but just sort of people buying groceries, coffee, beef, have problems coming up with the money. And when I would say coming up with the money, I don't mean coming up with the value or the wealth, literally figuring out the physical means or the credit of just making basic transactions.

And I think that what we find is century after century, people make up their own money. So you can tell them this gold standard is very good for you. It keeps money sound. But people kind of don't care. And when things go pear shaped, they don't appreciate the value of the gold standard, which is generating inflation.

They just make their own money. And often it's very creative money that works financially. It's not just randomly printing up tickets and saying, poof, this is money. A lot of times, there were hard times money during financial crises in the 19th century in America. A lot of cities printed up their own dollar notes during the Depression in America.

This is really recent. Within the last 100 years, people have been doing this. And so people will make up their own money at their own level to work no



matter what you do at the large international policy level. And I think that's really important and we need to pay more attention to that. So I think the one single thing, if I had anything to sort of say with the book, it's

that the central challenge of the dollar has not changed for 500 years, which is that big money dollars work. Small money dollars are always a challenge. And this comes from an insight from, as you point out, College of Pola. He was an Italian medievalist who was looking at medieval money and pointed out exactly what I just explained to you.

Brassage costs are higher on small coins, so men's prefer to make big coins. I read that and I thought, that's the Fed. That's policy now. That felt very familiar to me as a contemporary policy writer, in addition to somebody who's trained as a historian. Well, obviously the book does go into all of that about the kind of big money, small

money and how the uses of money and use of ledgers, et cetera. But it is primarily about the US dollar and I suppose the interesting feature of the story is how the dollar came along before the US as a country came along. So it wasn't as if they on day one said, we're going to have a currency called the dollar. The US adopted what was already a global currency. So maybe give us a quick sense on that evolution of the dollar prior to the founding of the

US. Alexander Hamilton, the father of American finance, called it the ancient dollar. It was very clear to him that it was an old coin, it was an old idea at the founding of America. I think one of the things that we learn in school about money is that there's this implicit assumption of sovereignty, monetary sovereignty, which is that there's a lot of hand waving

around the definition of this. But basically it's the control of country has over its own money. So the assumption is new country, new money. When you look at the founding of America, you end up with a problem, which is that there was a new country, they had some measure of political sovereignty, but then they chose as their currency what was essentially a Spanish coin from Mexican and Bolivian silver with

a German name. We would have expected that they would have called their currency the Washington, but they didn't call it the dollar. And I don't think that they borrowed a word from the German. They borrowed a money. They basically adopted an existing currency zone that was defined by a huge silver coin

that at the time was minted by Mints in the Spanish Empire, Mexico City and at Potosí, where there were silver loads in what is now Bolivia. When I pulled on that thread, I ended up literally at the bottom of a Czech silver mine, trying to figure out where this came from. And so the word dollar comes from a valley in what is now the Czech Republic that was settled by, settled by, that was, so the valley had silver in it.

The person who claimed ownership of the valley, who didn't even have real ownership of the valley was sort of a local account, it was a real dirtbag. Stefan Schlick is often sort of written down as the sort of the father of the dollar. And it turns out he just was never in control of anything, kind of owned the property in entirety with his brothers and several of his cousins, and just decided, well, once they discovered silver, that he was going to be the lord of the valley.

So the problem with silver in the ground is that it doesn't become silver in your hand without a bunch of really expensive industrial processes. And this was true in the 16th century as it is today. And so he realized that he needed capital and he needed expertise. So he brought Saxon investors and Saxon miners over the mountains to this valley in Bohemia, to illegally mine and refine silver.

They named the valley in their own language, Yacham's Tall, the valley of St. Yacham. The silver that came out of that valley was never meant to be coins. So usually, and particularly at the time, there was one centralized mint in Kutnehorah in Bohemia. And it was a way that the king could claim his own share of the silver, and the nobles who had always lent money to the king could immediately then get paid back.

The mint was not a way to create coins necessarily, it was a way to control the silver. But one thing that the mint did, the sort of standard centralized mint in Bohemia, was it created lots of big and small money, the full range

of coins that everybody would need for small transactions and big transactions. This mint in Yacham's Tall, because it was kind of a wildcat mint, it was outside the Bohemian system and it was illegal and for a while nobody knew what was going on.

Eventually, what they did was they just created high value silver coins that had one job only, which was to pay dividends to the Saxon Silver Investors. If you were a Saxon Silver Investor, you expected your mining company to work in a certain way. That's the way it already worked in Saxony. That's the way they did it in Yacham's Tall as well. So you got this big silver coin as your dividend, it was a very high value coin.

It was worth like a week of labor and I calculated 21 pounds of bacon. It was sort of not something that you could hand over a counter for some bread. It was really big coin. It clanks. In fact, I have one in my bag. When I was talking to Peter Vorol, who's a Czech historian, who's done a lot of the legal archival

research in early Yacham's Tall that I leaned on in the book, he gave me a full silver copy of one of these original dividend coins. I'm going to drop it on the desk here. It goes, it doesn't ring, it clanks. That's a heavy coin. And so because the mine only produced dividends, you had this incredibly high volume flood

of these big coins and they became useful as money for just merchants. So you get very early on. These coins are flooding up through Leipzig and into the Baltic with trade. They're passing by Sweden and Denmark. You get copies in Sweden and Denmark of this coin. Because it came out of the value of Yacham's Tall, it was called the Yacham's Tall.

And then you get Taller in German and then eventually Daalder in Dutch and then Dollar in America. So as German cities, Hamburg, Lubbock, are copying this, it gets copied in the Netherlands as well. And then it ends up in London. So within well less than a century, Shakespeare in two plays mentions dollars.

He mentions it in a Tempest and he mentions it in the Scottish play. And so to me, that says this thing called the Dollar, which sat well outside of the

British monetary or the English monetary system of pounds, shillings and pence, was familiar to groundlings at the Globe Theatre. They knew that it was a big silver coin that came from the Low Countries for English wool and English cloth.

So this coin spread because it was never meant to be local money. It was always meant to be a dividend for investors, which made it very valuable as what we think of now is high-powered money or what Carlos Polo called big money. It was money for merchants, for settling long distance trade. And so when the Spanish discovered that they had silver, first through plunder in Mexico, Charles V, the Holy Roman emperor, emperor, you know, in charge of Castile, but also brand

new vice royalties in Latin America, realized that even though he was emperor of all he surveyed, he could not control the silver markets. And so if he was going to sell his silver through Antwerp to the Portuguese who were bringing it to India and China, he had to put it in the exact same form that those silver merchants in Antwerp were already familiar with, which was this big silver coin that a century earlier had come out of a valley in Bohemia.

So even Charles V had to figure out how to hack his own monetary system to create something called a Real de Ocho, a piece of eight, an eight-real piece. But what it really was was a copy of this coin. Chipola said, with big money either, tends to be one dominant form of money, and a lot of other monies are what he called more or less faithful copies of that dominant money. So Charles V had to create a more or less faithful copy of the Jachamstaller, which had

become by then the Daughter or the Dollar. And in order to get his silver in the global markets, again, very clear to the English what this is, his son Philip II continued amending dollars in even greater volume. And we get in the 1570s a mention in English of a Phillips dollar. So this thing arrives, it's gone through this tortured path, it's silver that's coming out of this incredibly repressive forced labor mining system in Bolivia, but it eventually

gets minted by the Spanish and when it arrives in England, they immediately see it for what it is, which is a dollar. So by the end of that century, you have colonies in America that are on the edge of this vast international trading system in dollars, it's both in the Atlantic and in the Pacific economy, and the dollar becomes the dominant form of useful currency in America as well. So by the time of the founders, you end up with this system where they're takers of foreign

money, they're not producing domestic money, there's no domestic gold or silver in America, they don't even have any until the 1830s in Georgia. And then eventually, of course, the great gold strikes in California in the 1850s. So America adopted an existing global monetary zone, they did not create their own dollar zone. And so that to me is incredibly important because what it says to me is sovereignty is overrated

or rather, it's very important, but we tend to assume sovereignty when in fact, monetary sovereignty has to be fought for. You have to figure out how to take a currency and domesticate it and make sure it works for everybody. I don't think that process was complete with a dollar until 1932. We can talk about the transition from this silver dollar to bank money, but you know,

there are a lot of stumbles along the way, a lot of financial crises, a lot of experimental regulation to figure out how to create the system we have now. That silver dollar from Mexico was current as money in America until the 1850s. There was no moment where America said, we're on our own, time for our own money. It was a long process of adopting this foreign money and then figuring out how you use banks to produce your own money that's reliable for everybody.

As you're talking about the money for merchants versus money for everybody else, obviously there are those parallels with the use of the dollar at the moment in the global system. So the dollar is the reserve currency, so obviously we can get into your dollar market in a moment. It's used widely amongst merchants, but at the same time, it is the currency of day-to-day transactions in the US as well.

So I guess as that part of your point, that dual role creates some tensions or some interesting trade-offs. Absolutely. Yeah, absolutely. I think you're

right. Everyone else in the world has a dual currency, just not the United States.

I think it's really convenient. We talk about Europeans and Chinese politicians often complain about the power of the dollar, and I think there are problems with the system, but I also think it's kind of convenient to have your domestic monetary policy where you can manage things for yourself, and then this offshore system of dollars that you're not necessarily in control of, but for the Germans, for example, in the 60s and 70s, it was really handy to keep German wealth offshore in this

dollar system so that it didn't come home and cause inflation and deal with this and disrupt this model that Germany had of low-wage, high-value machine exports. And so I think the dollar system is really useful right now for a lot of people. I think it's good to have a global currency, an international currency that's used for international transactions that's reliable for everybody, that's got a predictable value that works for merchants, but also this more flexible system at home of a different kind

of money that works for people making purchases at the grocery store. These are two different kinds of monetary system and always have been. Let's take a step back and talk about that. Obviously, what is interesting, you wrote an article in the FT couple of months ago about the, there's no such thing as a petrodollar is what it was called, but it was about the petrodollar, but also about the evolution of the euro dollar markets, and in it you mentioned

how there's 14 trillion in offshore dollars and 19 trillion in domestic dollars. So the size of the US external dollar market, it was nearly as big as the domestic market, which is a fascinating statistic. And obviously, I mean, there was a whole bunch of different factors came together to, I suppose, initiate the development of the euro dollar market with kind of tacit approval from both the US and the Europeans.

And now it exerts a huge influence on the global market. And I guess the point of your article was that it's getting, maybe it's getting stronger. I made all the talk of the dollar as potential demise, that stability and the dominance and the influence of the euro dollar market is as great now as it's

ever been. Is that fair to say? So I think this is another inheritance that we have from economists that makes it difficult

to understand policy now. So when we learn that money is fiat, it makes us incurious about how it's actually produced. So the basic just-so story, and all histories are in some ways a simplified model, the basic just-so story that we learned about money, again, if we took a macro class or a micro class for that matter in college, is that money used to be silver and gold, and then it became paper, and then we took silver and gold away, and now it's fiat.

This is Latin for let it be done. The state says this is money, and then it's money. Poof. I think that betrays a profound lack of curiosity about how money is actually produced. When we look into how money works and how it's actually produced, I think we're able to see things that are hard to see with a framework of fiat.

And the most important part of that is what you just brought up, euro dollars. So in America, we obviously transitioned away from the system of using foreign silver coins, and what we did was we chartered a bunch of banks. Constitution said individual states can't make their own money, so the states chartered banks to do it for them. A bank note is essentially just a piece of paper that says, you know, the bank is giving

you a loan, but it's giving you a loan in the form of paper that you can spend as currency rather than silver or gold. It took a long time to make this model work, but by the 1950s, of course, in America, this model is pretty reliable. Not only is it reliable, it's the only reliable, large industrial economy left in the world. Not only that, gold is still important at this point in the financial system, America's

got 60% of the gold. It turns out, though, that the first thing that happened is that American deposits became useful for people abroad. What we would have predicted in the system, and what economists thought was going to happen, is that foreigners were going to sell things to America, which they started to do again in the late 50s and early 60s, and then they would earn American bank deposits as payment

for that, and then they would present those bank deposits at the Fed, which would go through a process that would eventually give them gold from Fort Knox. That didn't happen. They were really confused about it at the time, and so we've got this record of the Fed sending researchers over to London to try and ask, like, what are you doing with these deposits?

They're coming back and they're telling the Fed, you're not going to believe what they're doing over there. What they were doing was just trading claims on American deposits. If you had access to deposits in America, you could sell that to somebody else in London who could then use it as cash. This system was really convenient as we talked about for other countries.

It was convenient for Germany, so it could continue its export model. It was really convenient for America because nobody was showing up at the front door of Fort Knox and asking for gold, or far fewer people than would have been otherwise. Then a crazy thing happened, and again, the Fed had to send people over to London to figure this out, which is that banks in London start making their own dollars. If you believe in the fiat system, you do not have an explanation for how this happens.

What's going on, and Milton Friedman pointed this out in the late 60s, is banks were marking up their own ledgers with brand new loans, brand new deposits, denominated in dollars, and they were doing this in London. You have a dollar-denominated banking system anchored to deposits in American banks that's creating brand new dollars that are outside the reach of the Treasury or the Fed or anyone. It's this dollar system that defines global trade, I think.

If you believe in fiat, then you believe that money is a collective social institution, which it surely is, but the details matter. You believe that we're not quite sure why it is that everybody uses dollars, but everybody got into the habit of using dollars, so now everybody has to. I think a much more compelling description, a much more compelling story, is that once banks developed this offshore dollar system, it worked, and it still works today.

This is the challenge. If you're going to settle trade, you need a short-term bank loan to buy your inventory so that while it's on a ship moving to



somewhere else, you can have access to cash right now. That's all happening in dollars. That's all happening with bank loans in dollars inside this global dollar system that has slowly been figured out over the course of the last 50 years.

Very difficult to imagine what replaces that or how it gets replaced. Again, it's not a perfect system. There are lots of drawbacks for this system. I'm just talking about these institutions are very deep, and the way international trade works right now is you need a short-term dollar loan, and banks in China are doing this as well.

We've seen growth in euro dollars, meaning offshore dollars in China over the last several years in addition to other places in the world. When you look at those numbers, they're continuing to grow. When we worry about the loss of dollar dominance, we have to look at the dominant form of dollars in the world. It's euro dollars.

Man, that doesn't seem to be decreasing. We should ask ourselves, why isn't it decreasing? It can't be because everybody in the world is under a collective delusion. It must be useful for them. It must be working. Is that just a network effect at this stage?

Obviously, there's various reasons why it's coming to existence. The point you make in the book is it's not necessarily just underpinned by US military might, which other commentators have made that point. Why does the dollar mean ... Obviously, it's here now, so everybody use it, but what is it underpinned by? I think if you just say it's a network effect, then you think of it as fragile.

You realize that everybody agreed to do this, and then you think transition away from the dollar is just a collective action problem. Everybody has to agree to use something else. I think there are three things that anchor the value and the use of the dollar internationally right now. The first is America really went through a ton of financial crises and a ton of experimental

regulation to figure out how to get all of its banks to ensure their deposits. We come out of the Great Depression with a national federal deposit

insurance scheme. Nobody else has this. There's no European-wide Eurozone deposit insurance. Every country is responsible for their own deposits on their own. That means there's this vast pool of safe, regulated, insured deposits.

This is just incomparable still today to what anybody else has in the world. In addition to that, the Fed figuring out slowly over the course of the 70s and 80s, and then implementing it in the 2000s and the 2020s, it offers what we call swap lines. This is just short-term loans that the Fed makes to other central banks. Those other central banks then make short-term loans in dollars to their own commercial banks to rescue them in a crisis.

After the crisis, this whole thing gets unwound and the swap lines go back down to zero. This is an incredibly effective way to make sure that dollars always have value. There are, you want, denominated swap lines that People's Bank of China offers. They're not nearly in the volume that the Fed offers. Other countries have come to realize that those swap lines come with strings attached. If we're looking at the three legs of the stool to give the dollar value, there's this

vast pool of insured deposits, there's the swap lines, and then the third thing is hard to define. That's just institutional quality of the United States. We do have pretty good bank regulation. I have some notes, but it kind of works. If we worry about the dollar, we need to worry about these three things.

There doesn't seem to be a challenge to deposit insurance, although I worry about stable coins because it's basically a form of bank dollar that does not have insurance. The stable coins grow, I think we might see a challenge to that. I think the more likely outcome is that we'll get a crash in stable coin value and they'll be brought into the regulatory fold a little more tightly. Stable coin providers will have to pay for deposit insurance because we just can't have

people running around with things that claim to be dollars that are not absolutely secure. Devin Warsh with the Fed has made some noises that he would like to take another look at the swap lines. He has said in testimony and then in follow-up letters to the Democrats and the Senate that he believes that domestic monetary policy should be independent from the

Treasury of the White House, but that's not true once you cross the border, that international policy

should be coordinated. We could see, though he has not explicitly threatened this, we could see a world in which dollar-denominated swap lines from the Fed have political consequences, they come with strings attached. To have access to them, you need to be doing something America likes, whereas in the past, all you had to be was a responsible central bank.

If the Fed had good relationships with your central bank, then you could get dollars in a crisis. Those swap lines tended to expand during crises. It's a pretty good system, and it doesn't have any political strings attached. They are not carrying out American policy, they are rescuing the global dollar system. Which is a weird question.

It's interesting. It's just going to say in terms of the swap lines, obviously it's been the crisis periods where we've seen the implementation and extension of the old swap lines, so 2008 and then again 2020. Those episodes have highlighted, I guess, this vulnerability of this global system with euro dollars, where there are a lot of entities with dollar liabilities outside US, so in

the times of stress when people are scrambling for dollars, there is this spike in the value and this kind of scramble to get dollar liquidity. In the absence of the swap lines, it does create potential conditions which could be, I guess, harmful for the US. That's why they provide the swap lines in the first place. Yeah, absolutely.

The reason for that is that if a global economy collapses, surely the American economy will, too. I'll buy it, but they're not carrying out American policy in the sense that the Treasury also has its own swap lines, distinct from the Federal Reserve swap lines, and Scott Bassin has made very clear that he's going to use those swap lines for political purposes. He's going to dangle them for Argentina to encourage Argentinians to carry out policies

that are friendly to America. That's not been how the Fed worked in the past. They basically just said, if you've got dollars, we're going to make

you whole on your dollars so long as you're a responsible economy with a responsible central bank. You talk about the strings, and I think the swap with Argentina came from the Treasury, not the Fed.

It did. I mean, as strings attached, is this as consistent with a more fragmented world where you have a dollar block, maybe a yuan block? Is that just the direction of travel or how easy is it? Yeah, I think it's possible. I don't want to say this is immediately happening, but if we actually follow this logic, it's

possible that countries may need to decide in the future whose strings do I prefer, the PBOCs or the Fed's strings. If indeed, the swap lines become politicized, as we fear they could be, but don't yet know that they will be, yeah. Countries would have to decide in a way that I don't think they have to decide right now. But I think the other pillar in the value of the dollar is just the general institutional

quality of the United States. I think there are some questions there right now. The problem with that one is that it's really hard to define, and we can walk through deposit insurance. We can walk through swap lines. These are very specific financial arrangements that do not rely on a network effect.

They are arrangements that help us explain why people make rational choices to use the dollar and not just out of habit. The institutional question, I don't know, I can't define it for you. I mean, I can define it, but I can't measure it. But I think that there is some question abroad about the quality of American institutions. So far, really seem to have done that much damage to the global use of the dollar.

So it seems like people abroad are looking at the quality of American bank regulation and the reliability of those swap lines before they're looking at sort of the general quality of the institutions of the United States. I've been telling people recently, we can't think of the dollar as a referendum on the United States. People are mad at the United States, but they're still gritting their teeth and using the dollar.

There's got to be reasons why. Again, I don't say that as an American. I don't say that with disdain. I can understand the frustration that people

have, but I just think that we've got to get away from this idea that people use the dollar because they love and trust America. It turns out that people abroad, when you travel abroad as American, people will tell

you what they feel about America, and it has changed. I was an exchange student in rural Germany in the 1990s. They loved Americans, could not get enough of Americans, so happy to talk to me. That's definitely changed wherever I go in Europe. And yet, the dollar system endures, and I think we have to figure out why, and I think it's those other two pillars, the reliability of American bank deposits and the swap lines.

I also think that over the last 50 years, America has become really good at connecting investors to projects that need investment. This old historians joke about how Rome disappeared and all that was left was a church, England's empire disappeared and all that was left was a bank of England. It's possible, though I am not predicting it, that if America, in its current shape, ceases to exist, we could still have the New York capital markets.

People come from all over the world to raise money in dollars, and people come from all over the world with wealth to invest their money in dollars. There's a lot of basic understanding in knowing by doing that happens in New York to generate dollar-denominated assets that people can invest in. One of the reasons why the Eurodollar system was so strong in the 60s and 70s was that there was this leftover knowledge in London, in the city of London, in doing exactly that.

You have these two historical financial centers that are used to doing this, that continue to do it today, produce all this value in dollars. I think one other thing that you'd have to recreate somewhere else is, what's the Chinese financial center that can take absolutely anything and turn it into a USD-denominated asset? I teach at Princeton now.

My students are headed to Wall Street every year. That's their job, generate dollar-denominated assets. I try to grab them by the shoulders on their way out the door and say briefly, please don't cause a financial crisis. That's where they're headed. You're just transitioning to a related point.

You've been talking about the role of the Fed in all of this, the swap lines, but taking a step back, obviously you've covered the Fed for 20 years as a journalist. What people forget, I guess, about central banks is that they are banks like every other bank. That's part of the history. The Bank of England was a commercial bank too, and then it was a special commercial

bank financing the wars, *etc.* The Fed effectively is similar in that it is a bank like any other, but with special characteristics and qualities. But it's involved as well over time, and I guess that's been part of the Warsh concern about maybe Fed overreach, *etc.* We had QE and kind of expressed some skepticism of QE, and then in COVID we had the Fed extending

that to credit markets and considering a wider range of purchases. What's your perspective on the Fed now? Are you sympathetic to that idea of Fed overreach that Warsh talks about, or do you think they've been successful in managing a complex system? I think it's really important to think of the Fed as a bank and central banks as banks. When we talk about policy, often we sort of get abstracted into the Fed as raising or

lowering interest rates. What the Fed is actually doing is putting things on its balance sheet or selling things off of its balance sheet. That's a really flexible tool, and the tool of a central bank has changed over time. The model of all modern central banks is the Bank of England. It was a commercial bank that was chartered to help prosecute the war against France.

That was its job. Now, that's obviously not its job anymore, but relatively recently, the Fed decided that its job was to help prosecute the war against the Germans. The Fed bought federal debt to keep the price of debt down. It explicitly understood that that was its job during the war, and then it stopped after that and declared its independence.

Things change, things happen, and our central bank should be responsive to the needs of a democracy. I absolutely agree with central bank independence that on a day-to-day basis you should be free to make your own decisions as a central bank or free from party politics. However, the central bank is a bank that's chartered by, in America's case, Congress. If

we want to change the goal of the central bank, the mission of the central bank, we

should have the right to do that. I think it's interesting that Congress decided in the 1970s that the central bank was that the Fed's job is not just to manage full employment and stable prices, but part of that is also that it's supposed to help grow monetary and credit aggregates in a way that increases productivity. It's just forgotten about that. It doesn't like that's part of its job, because that part of its job means that it needs to

show discretion, which kinds of monetary and credit aggregates are the more important ones, which ones create productivity, which ones don't. That's a lot of decision-making, and it's a lot of politics, and the Fed doesn't want to be a part of it. They basically just decided, we're not going to do that, and so they don't. The Fed makes up its own homework, grades itself on its own homework, and then shows

up at Congress twice a year and says, look, didn't we do a good job? I think that, for example, if Congress wanted to say, the Fed has to help finance the green transition. I don't have any problem with that. There are a lot of central bank purists who will say, that's not the job of a central bank, and my answer is, look, none of this was the job of a central bank.

We had to invent central banks to do jobs that we needed them to do, and then we changed those jobs periodically, and maybe this is a job we needed to do. I'm sure you have listeners who are busy looking at my email to yell at me now, but I think it is important to think of the central bank as a bank, and to think about that balance sheet structurally and what it can do, and I think the Fed has gotten out of the habit of thinking that it's a balance sheet that can do anything other than buy treasuries,

and I think the utter reluctance by the Fed to ever contemplate buying anything other than treasuries is a profound failure of imagination, because in a crisis, they sure do jump in and save markets, but they won't do that during normal times. We just don't think of that, and I don't think that's a legal challenge, or I think that's a philosophical challenge. The Fed would need the cooperation with the Treasury to buy anything other than agency

securities and treasuries. Congress would need to give the Fed a new job. Stuff would need to happen, but the barrier to that is this rock-solid assumption that we've inherited from economists that the Fed has one job, we have perfected the job. It cannot develop any farther, and we can't ever stray from this single job of pushing one interest rate up and down to influence inflation and employment.

I think we need to think really structurally about what those tools are, whether they're working, they aren't right now, by the way, and then what they could do potentially in the future, and to do that, we have to think of central banks as banks. It sounds like all of that is up for review now, I mean, based on everything we heard from Kevin Warsh, these various task forces, et cetera, so I mean, you think we are on the coast, but over potentially different looking Fed for the Nixon reviews?

Yeah, absolutely. Look, I think Kevin Warsh is right about QE. The Federal Reserve decided after the financial crisis that it was going to do, what it had failed to do in the 1920s, Ben Bernanke famously is a financial historian. He, of course, knew that Milton Friedman had accused the Fed in the 1930s of failing to buy a bunch of treasuries and increase money supply, and he said, well, I'll be damned

if I all fail to do that this time. So he did it. I also think that just buying a lot of treasuries is politically convenient for the Fed, which is they can't get in trouble doing it because it doesn't favor one political party or another. It does, of course, have massive distributional consequences because if you have existing assets, it can become more valuable when the Fed buys much treasuries, but they do that

not because it is the most effective thing to do, but because it is a thing that they can do that they don't think they'll get in trouble for. The Fed is unwilling to take any risks that will get it in trouble, and so I think that QE is a default policy that doesn't really work that well. It basically, it pushes off the value of existing assets. That's all it does.

It makes wealthy people wealthier. I'm not sure that's a policy goal we want in America. And so the Fed could have, and it had the legal authority to during the financial crisis, buy a bunch of mortgages, but that would have



been a lot of fiddly boring work. It would have had to renegotiate these mortgages with people one by one. It would have had to have hired the administrative staff to do that, but it would have been

effective. And instead, what we got was a lot of people turned out of their homes, which is incredibly politically scarring and economically scarring. So I think when you cover the Fed, it's very easy to be captured by the Fed. They're lovely people. They all have PhDs.

They clearly want what's best for America. And if you've covered Washington, DC, you will know that all three of those traits are rare, particularly in one human being. So it's easy to sort of, when you cover the Fed, want them to flourish, fail to be critical enough of what they're doing precisely because they're such lovely people and smart people. But I think that we have to think really hard and critically about how the Fed failed to

use its balance sheet in the financial crisis. To do something that would have been really effective and instead did something by treasuries, which was, I think, explicitly a choice to prevent them from getting in trouble. And I'm not sure why the balance sheet is so large. I think it's large out of habit. It's very difficult to actually make the balance sheet smaller once banks get used to having

the balance sheet that big. Once banks get used to having that much in reserves rattling around. And so Kevin Warsh is right to say, what are we doing with this balance sheet? Why can't we ever shrink it ever again? Are we stuck forever with the ample reserves regime? Those are really good questions.

And in the past, the past when the Fed has done these policy reviews, it has absolutely failed to ask really hard questions about what its tools are and whether they're working. I think it's important that we're having that process. Okay. Okay. Interesting stuff.

We're just up against time. We do like to ask our guests before we wrap up for any advice for people who are looking to learn more about economics, economic history in your case, or things you've read, things you've done.

Obviously, you've gone back and done a PhD. So you've had that luxury of, as you say, reading a lot of books and stuff.

Any advice for people who are interested in economics and economic history? Yeah. There's a book I always recommend, which I think I mentioned earlier, which is Rebecca Speng's *Stuff and Money in the Time of the French Revolution*. You may not think you're interested in the French Revolution, but just the process of trying as hard as you can to figure out exactly how people paid each other and then exactly

how people manufactured money is a really interesting exercise and a really valuable one. Absolutely read that book. That's my one recommendation I give to everyone. Good stuff. Well, great to have you on, Ben, and fascinating to hear about the evolution of the dollar.

And obviously, if people are interested, they can go and find your book. It's called *The Almighty Dollar, 500 Years of the World's Most Powerful Money*, which I'm reading myself at the moment, and it's a great read. And obviously, you're at Princeton University at the moment, so I guess we'll hear more from you over time. From all of us here at Top Traders Unplugged, thanks for tuning in.

We'll be back soon with more content.

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